

Finley Desai

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Education

Stanford University

M.S. in Electrical Engineering, Coterminous Program – Incoming Fall 2026

B.S. in Electrical Engineering, Minor in Computer Science

GPA: 3.99/4.0

Palo Alto, CA

Expected Jun 2028

Expected Jun 2027

Selected Coursework: Digital System Design, Circuits I-II, Signals and Systems I-II, Computer Systems, Machine Learning, Feedback Control Design

Technical Experience and Projects

Stanford Tambe Lab

Research Intern, Robotics Chip-on-Board Module

Palo Alto, CA

Jun 2026 – Present

- Designing schematics and layout for a ~16-layer high-speed PCB in Altium for robot inference, integrating ~50 ICs, including wire-bonded devices.
- Developing the controlled-impedance stackup, microvia/blind-via architecture, and DDR3-speed routing constraints using pre-layout signal-integrity analysis in MATLAB Parallel Link Designer and Signal Integrity Toolbox.

Stanford Accelerate Lab

Research Intern, Hardware Accelerator Test PCB ([GitHub](#))

Palo Alto, CA

May 2026 – Jun 2026

- Designed a high-speed test PCB in KiCad for a custom hardware accelerator, integrating JTAG and peripheral interfaces and optimizing clock routing for operation up to 800 MHz.

Stanford Magnetic Resonance Systems Research Lab

Research Intern

Palo Alto, CA

Jun 2025 – Jan 2026

Custom High-Res Wireless Quadrature Digitizer PCB for MRI Radar Prototyping

([Conference Paper](#) | [GitHub](#))

- Designed a dual-channel quadrature digitizer with high-resolution ADCs and analog front-end circuitry for continuous-wave radar based vital signs monitoring in MRI settings; measured an ~80 dB improvement in dynamic range over SDR-based acquisition.
- Developed MicroPython/PIO firmware on a Raspberry Pi Pico W for high speed synchronized data acquisition and Bluetooth streaming.
- Created schematic and multilayer PCB layout; stuffed and tested the board using oscilloscopes and signal generators.
- Analyzed radar signal data using singular spectrum analysis to isolate respiratory and cardiac motion.

Stanford Molecular Imaging Instrumentation Laboratory

Research Intern

Palo Alto, CA

Mar 2024 – Jun 2025

- Developed and executed custom Monte Carlo physics simulations, analyzing large datasets to design and validate inter-crystal scatter correction algorithms and event-positioning strategies for PET imaging.
- Two conference papers accepted for presentation at IEEE Nuclear Science Symposium & Medical Imaging Conference (NSS/MIC); abstracts published in conference proceedings: [Event Positioning](#), [Inter-Crystal Scattering](#).

Bare-metal Systems Programming – Mango Pi RISC-V

- Implemented bare-metal C libraries and drivers for I/O, memory management, interrupts, keyboard input, graphics, printf, and a terminal interface on a Mango Pi RISC-V board.

Additional Experience

American Industrial Partners (Industrial-focused Private Equity)

Operations Intern

New York, NY

Jun 2026 – Present

- Built an inventory-aging model for Global Cellulose Fibers (GCF) woodyards to track wood quantities and types over time and estimate pine-chemical losses; presented model outputs to mill and woodyard managers.
- Investigated maximum sustainable rate calculations and reporting across GCF mills; designed a procedure to standardize the process and built financial models supporting the operating agenda and new initiatives.

Technical Skills

Language/Tools: Verilog, C, C++, Python, MATLAB, Altium Designer, KiCad, LTspice, Git, CMake, Linux

Hardware & Lab: PCB design & assembly, soldering, oscilloscopes, logic analyzers, signal generators, spectrum analyzers